9.4. Outliers in Data

Previous Section:

[9.3. Covariance and Correlation in Data]

Real-world datasets often contain observations that are very different from the rest of the data. These unusual observations are called **outliers**.

Detecting outliers is an important preprocessing task because they may significantly influence statistical analysis and machine learning models.

In this section, we will learn what outliers are and how to detect them using two classical statistical methods: **Z-Score** and **Interquartile Range (IQR)**.

We will create a dataset containing **200 students** with their **Height (cm)** and **Weight (kg)**. Approximately **20% of the samples are intentionally generated as outliers** to demonstrate different detection techniques.

```
import numpy as np
import pandas as pd

np.random.seed(42)

n = 200
outlier_ratio = 0.20
n_outliers = int(n * outlier_ratio)
n_normal = n - n_outliers

# Normal samples
height_normal = np.random.normal(170, 8, n_normal)
weight_normal = 0.45 * height_normal - 5 + np.random.normal
(0, 4, n_normal)

# Outliers
```

```
height_outlier = np.random.uniform(120, 220, n_outliers)
weight_outlier = np.random.uniform(25, 130, n_outliers)

height = np.concatenate([height_normal, height_outlier])
weight = np.concatenate([weight_normal, weight_outlier])

df = pd.DataFrame({
    "Height": np.round(height, 1),
    "Weight": np.round(weight, 1)
})

df = df.sample(frac=1, random_state=42).reset_index(drop=True)

print(df.head(10))
```

1. What is an Outlier?

An **outlier** is a data point that differs significantly from most of the observations in a dataset.

Outliers can occur because of:

- Measurement or recording errors
- Data entry mistakes
- Rare events
- Naturally occurring extreme observations

Outliers are important because they may:

- Skew the mean and variance.
- Affect correlations between variables.
- Distort regression models.
- Reduce the performance of machine learning algorithms.

However, not every outlier is an error. Some outliers represent valuable information, such as fraud detection, equipment failure, or unusual customer behavior.

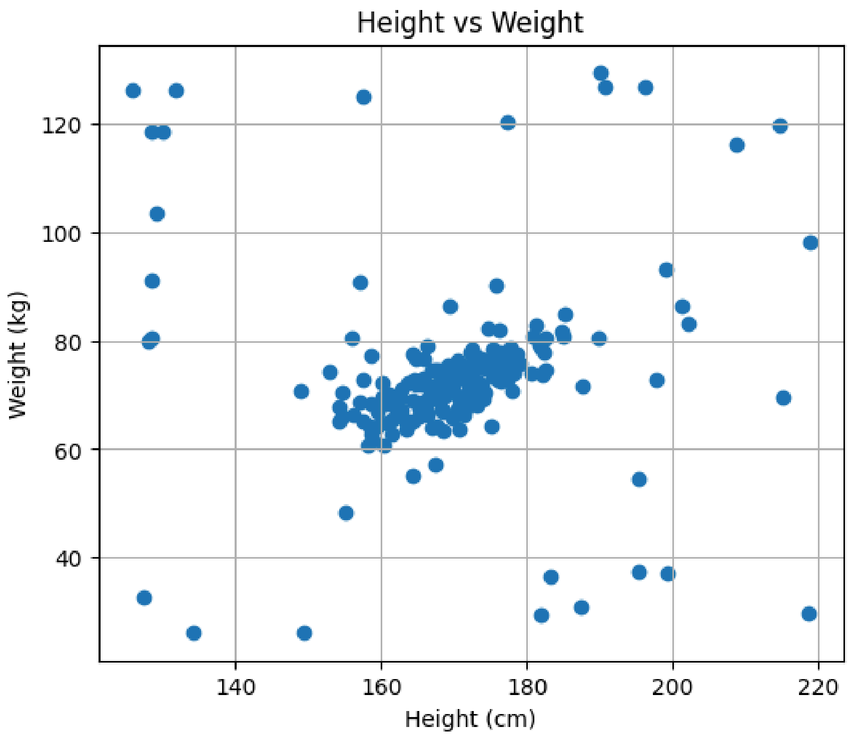
💡 ***Note:*** *Outlier detection may be performed before or after feature scaling. In practice, many machine learning workflows perform outlier detection after scaling for greater consistency.*

- Visualizing the Dataset: A scatter plot quickly reveals observations that appear far away from the majority of the data.

```
import matplotlib.pyplot as plt

plt.figure(figsize=(6,5))
plt.scatter(df["Height"], df["Weight"])
plt.xlabel("Height (cm)")
plt.ylabel("Weight (kg)")
plt.title("Height vs Weight")
plt.grid(True)

plt.show()
```



2. Detecting Outliers

There are many techniques for detecting outliers. Two of the most widely used statistical approaches are: Z-Score and Interquartile Range (IQR)

Method 1: Z-Score

The **Z-Score** measures how many standard deviations a value is from the mean.

$$Z = \frac{x - \mu}{\sigma}$$

where $x =$ data value; $\mu =$ mean; $\sigma =$ standard deviation

A common rule is: $|Z| > 3$ → The observation is considered an outlier.

- Example

```
from scipy.stats import zscore

z_scores = np.abs(zscore(df))
# We use max to take the worst outliers, either the height
or weight
df['z_scores'] = z_scores.max(axis=1)

outliers = (z_scores > 3).any(axis=1)
df_zscore = df[outliers]

print("Number of outliers:", len(df_zscore))
print(df_zscore.head())
```

Number of outliers: 9

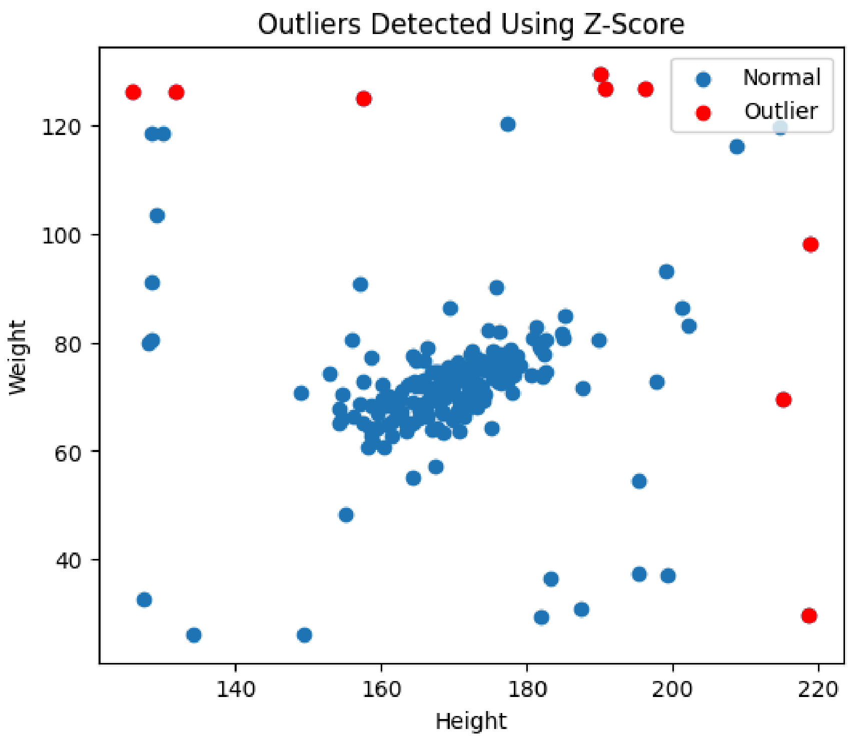
- Visualizing Detected Outliers

```
plt.figure(figsize=(6,5))

plt.scatter(df["Height"], df["Weight"], label="Normal")
plt.scatter(df_zscore["Height"],
            df_zscore["Weight"],
            color="red",
            label="Outlier")

plt.xlabel("Height")
plt.ylabel("Weight")
plt.title("Outliers Detected Using Z-Score")
plt.legend()

plt.show()
```



Method 2: Interquartile Range (IQR)

The **Interquartile Range (IQR)** measures the spread of the middle **50%** of the data.

$$IQR = Q3 - Q1$$

$$\text{Lower bound} : Q1 - 1.5 \times IQR; \quad \text{Upper bound} : Q3 + 1.5 \times IQR$$

Any observation outside these bounds is considered an outlier.

```
Q1 = df.quantile(0.25)
Q3 = df.quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = ((df < lower) | (df > upper)).any(axis=1)
df_iqr = df[outliers]

print("Number of outliers:", len(df_iqr))
print(df_iqr.head())
```

Number of outliers: 33
Notice how the number of outliers are far different from using Z-Score

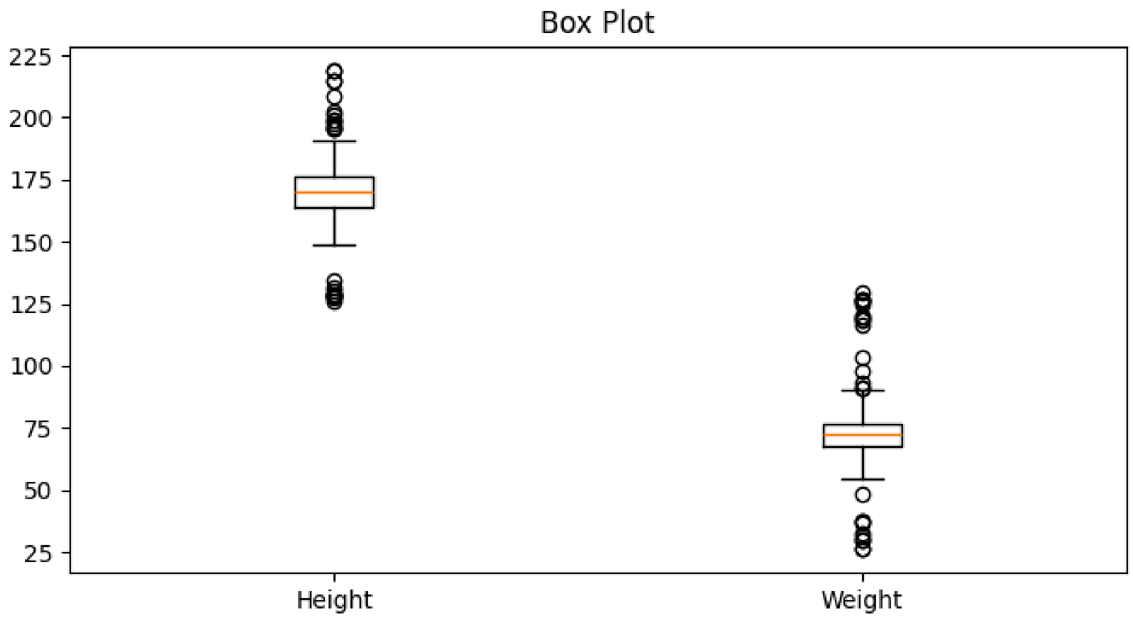
- Visualizing with a Box Plot

```
plt.figure(figsize=(8,4))

plt.boxplot([df["Height"], df["Weight"]],
            labels=["Height", "Weight"])

plt.title("Box Plot")

plt.show()
```



The points outside the whiskers represent the detected outliers.

Advanced Methods

Besides statistical approaches, modern machine learning provides more sophisticated techniques for anomaly detection, including: Isolation Forest; Local Outlier Factor (LOF); One-Class SVM; DBSCAN

These methods are particularly useful for high-dimensional datasets and will be discussed in a later section.

Summary

In this section, we learned how to identify unusual observations in a dataset.

- An **outlier** is a value that differs significantly from most observations.
- Outliers can be caused by errors, rare events, or genuinely unusual cases.
- Outliers may distort statistical analysis and machine learning models.
- **Z-Score** detects outliers based on the distance from the mean using standard deviation.
- **IQR** detects outliers using quartiles and is less sensitive to extreme values than the Z-Score.

- More advanced anomaly detection algorithms, such as **Isolation Forest**, are commonly used for complex datasets and will be covered in a later section.

References

https://www.geeksforgeeks.org/machine-learning/what-are-outliers-in-data/

https://www.geeksforgeeks.org/machine-learning/z-score-for-outlier-detection-python/

https://www.geeksforgeeks.org/machine-learning/interquartile-range-to-detect-outliers-in-data/

Next Section:

📊 9.5. Evaluation Metrics